fischertechnik
Old School Challenge

Briefing Book

Fischertechnik old school challenge

fischertechnik is a German construction and educational system introduced in the 1960s by Professor Artur Fischer, the inventor of the famous Fischer wall plug. Unlike many construction toys that focus primarily on creative building, fischertechnik was designed from the outset as an engineering and technology learning platform. Its unique modular building blocks, gears, axles, bearings, and structural components allow users to construct not only static models, but also functional machines that demonstrate real-world engineering principles. Over the decades, the system has been used by hobbyists, schools, universities, and industrial training programs to teach concepts ranging from basic mechanics to advanced automation.

The early Hobby series, culminating with Hobby 4, introduced users to increasingly sophisticated mechanical systems. Builders could explore structural engineering, gear trains, transmissions, camshafts, crank mechanisms, steering systems, and motorized machines. The addition of electrical motors, lamps, relays, and switches expanded the learning experience into electromechanics, enabling the construction of automated devices that could react to inputs and perform controlled sequences of operations. These models closely mirrored the technologies used in industrial machinery and manufacturing equipment of the era.

The challenge described in this document focuses on the first generation of fischertechnik electronic control systems. Participants will work with classic analog and digital logic modules, allowing them to create machines capable of very basic but challenging constructs. These modules represent an important technological milestone, bridging the gap between purely electromechanical systems and modern computer-controlled automation. While later generations of fischertechnik expanded into pneumatics, robotics, programmable controllers, and computer-based automation, those technologies fall outside the scope of this challenge. Our focus remains on the fascinating transition from mechanics and electromechanics to early digital logic, where complex machine behavior could be achieved using only gears, switches, relays, and a handful of electronic building blocks.

And yes, this paragraph has been built with AI, using very carefully crafted prompts by an experienced and passionate fischertechnik never growing up 10 years old (in his head) who feels the same as the same day when he first touched the iconic brick. You can likely crack this challenge entirely with GenAI. But, don’t be an ass, imagine you are a 10 years old in the 70s and go for it, rediscover a world of reading the instruction, learning and crafting yourself with the components. This briefing book should have all the information you need.

- d4e5, aka dimiexter, aka d7

The Fischertechnik Master Flag Challenge

Challenge: The Fischertechnik Master Flag

You have been provided with a collection of vintage fischertechnik components, including structural parts, gears, motors, relays, switches, analog electronic modules, digital logic modules, and a pre-built "Fischertechnik Master Flag.” The Fischertechnik Master Flag features a digital logic circuit built around a NAND gate. When all required input conditions are met simultaneously, the logic circuit will trigger a control mechanism that raises the master flag. The machine exposes four input terminals and has a test mode and a raise the flag mode (for you to practice). Your objective is to design and construct four independent systems using the provided fischertechnik components. Each system must generate the appropriate signal for one of the four NAND gate inputs. You’ll receive a flag for each on of the 4, so this challenge has a total of 5 flags. Flags are handed manually by a human upon completion of each steps.
The four required constructs are:

1. Manual Input: Construct a simple push-button circuit. When the button is pressed:

- A small indicator lamp should illuminate.
- The first NAND gate input should be asserted.

2. Electromechanical Monostable: Construct an electromechanical device that behaves like a monostable (one-shot timer) using only mechanical and electromechanical components. The system should use components such as an electric motor, gears, cams, switches, relays, and indicator lamp.

Required behavior:

- The system starts idle with the motor stopped and the indicator lamp off.
- Pressing a start button initiates a cycle.
- The indicator lamp turns on during the active portion of the cycle.
- At the end of the cycle, the indicator lamp turns off.
- The system returns to its initial state and is ready to run again.

The indicator lamp shall drive the second NAND gate input while the cycle is active.

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3. <u>Analog Light Detector</u>: Construct a light-sensitive detector using the provided analog electronic components.

Required behavior:

- A light-sensitive cell detects illumination.
- Detection activates a lamp.

The indicator lamp shall drive the third NAND gate input.

4. <u>Magnetic Detector and Monostable</u>: construct a magnetic detection circuit using the provided analog electronic modules.

Required behavior:

- A magnet triggers the detector that activates a monostable circuit.
- The monostable turns on an indicator lamp for a short period of time.

The indicator lamp shall drive the fourth NAND gate input.

Objective

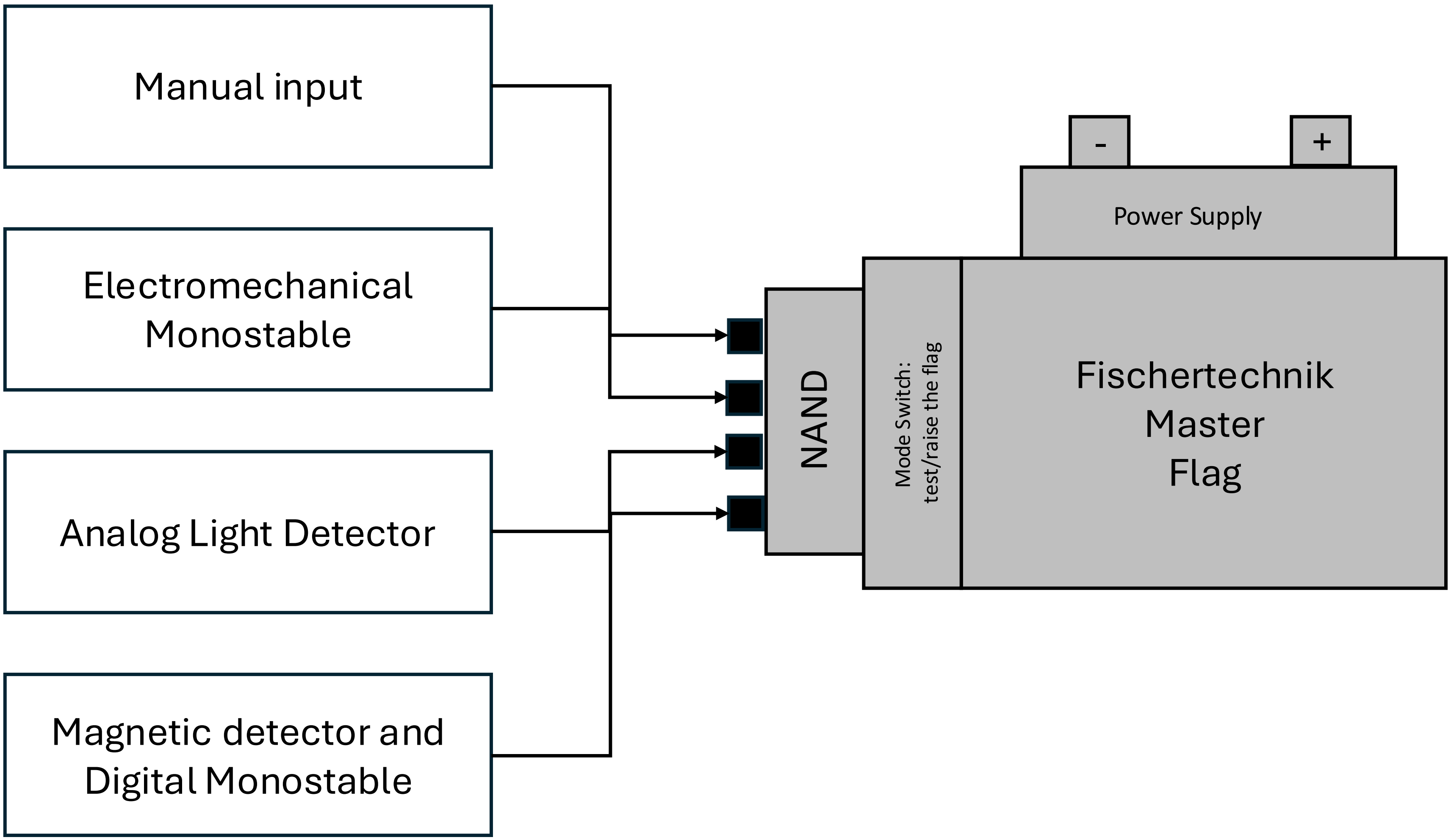
Once all four constructs have been completed and connected to the Fischertechnik Master Flag, you must determine how to activate them so that all four NAND inputs are asserted simultaneously. The machine is one 'test mode' initially. Once you are confident you can raise the master flag, flip the switch to 'raise the flag' mode.

Rules

- You may use any of the provided fischertechnik components.
- You may modify and rebuild your constructs as often as necessary during your session.
- You may take notes, photographs, measurements, and sketches.
- You may not modify the Fischertechnik Master Flag itself.
- The challenge is timed in 20-minute sessions.

If you do not complete the challenge during a session, document your progress and return during a later session to continue your work. Good luck, and welcome to the world of mechanical, electromechanical, analog, and digital automation.
More details on how to work on this flag next page.

Block Diagram



How to work on this flag during the CTF

You have been given this briefing book when you signed up for the CTF.

The schedule is as follow:

- **Mornings**, you have free access to a bunch of fischertechnik parts to familiarize yourself with the system, understand and test the components. You can then sketch your solution, but you will not have all the parts to complete the challenge
- **Afternoons**, from 1pm to 6pm, 20 minutes sessions at the top and bottom of the hour. During these sessions, you can build your solution. If you do not finish, you can come back for another session (restarting from scratch). You cannot do two sessions in a row; you need to pass a least one session (and go to the bottom of the queue if there are other teams in line after you)

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